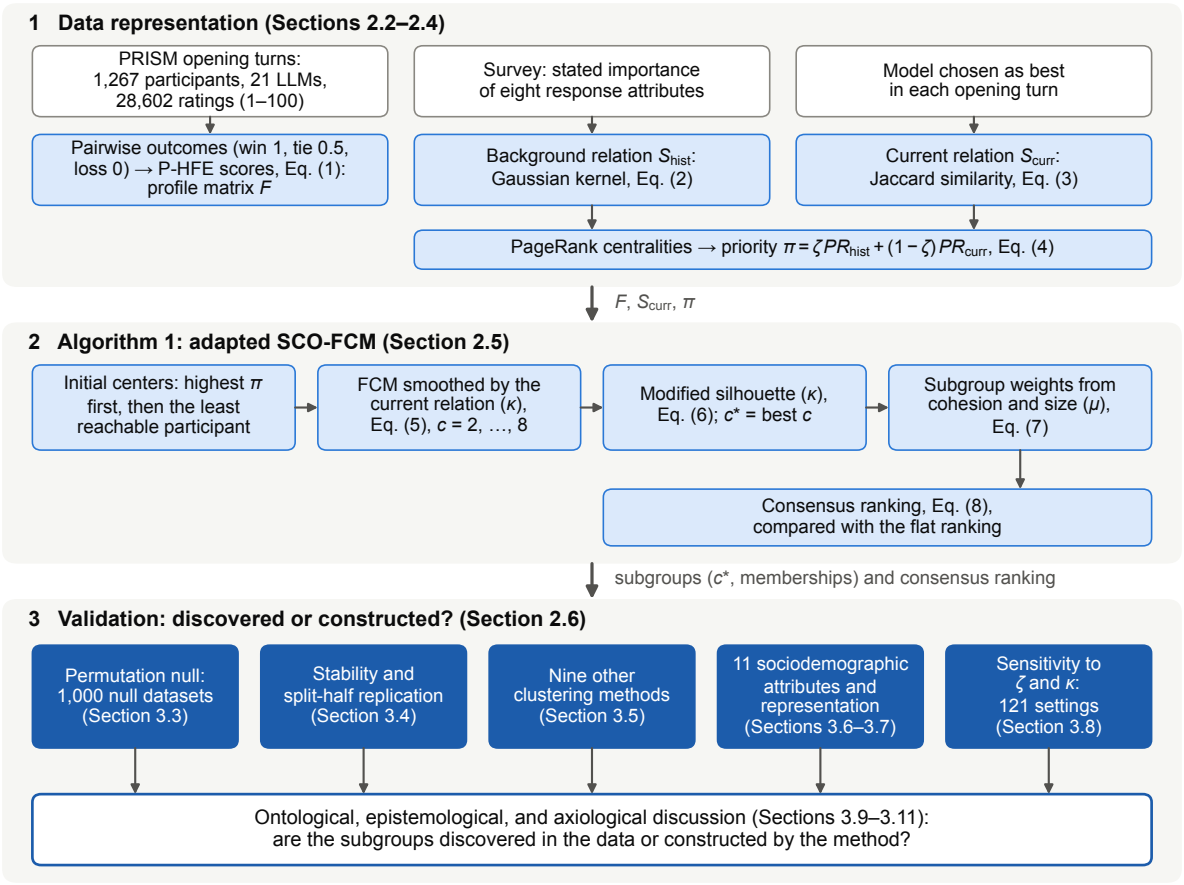

1 Data representation (Sections 2.2–2.4)
PRISM opening turns: 1,267 participants, 21 LLMs, 28,602 ratings (1–100)
Survey: stated importance of eight response attributes
Model chosen as best in each opening turn
Pairwise outcomes (win 1, tie 0.5, loss 0) → P-HFE scores, Eq. (1): profile matrix F
Background relation S_hist: Gaussian kernel, Eq. (2)
Current relation S_curr: Jaccard similarity, Eq. (3)
PageRank centralities → priority π = ζ PR_hist + (1 − ζ) PR_curr, Eq. (4)
F, S_curr, π
2 Algorithm 1: adapted SCO-FCM (Section 2.5)
Initial centers: highest π first, then the least reachable participant
FCM smoothed by the current relation (κ), Eq. (5), c = 2, …, 8
Modified silhouette (κ), Eq. (6); c* = best c
Subgroup weights from cohesion and size (μ), Eq. (7)
Consensus ranking, Eq. (8), compared with the flat ranking
subgroups (c*, memberships) and consensus ranking
3 Validation: discovered or constructed? (Section 2.6)
Permutation null: 1,000 null datasets (Section 3.3)
Stability and split-half replication (Section 3.4)
Nine other clustering methods (Section 3.5)
11 sociodemographic attributes and representation (Sections 3.6–3.7)
Sensitivity to ζ and κ: 121 settings (Section 3.8)
Ontological, epistemological, and axiological discussion (Sections 3.9–3.11): are the subgroups discovered in the data or constructed by the method?